



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment - 3

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Subject Name: System Design

Subject Code: 23CSH-314

Aim:

To design a **Social Media Platform** that allows users to register, login, create posts, follow other users, and interact with posts through likes and comments, ensuring high availability, scalability, and low latency.

Objectives:

1. To understand the working of a Social Media system
2. To identify **functional requirements** of the system
3. To identify **non-functional requirements** such as performance and scalability
4. To design a high-level system flow using **draw.io**
5. To understand core entities involved in the platform

Procedure-

1. Identify functional requirements of a social media platform.
2. Define non-functional requirements such as scalability, latency, and availability.
3. Analyze CAP theorem trade-offs for social media systems.
4. Identify core entities required for system implementation.
5. Design the system architecture using Draw.io.
6. Validate the design against real-world social media behavior.

Functional Requirements -

1. Users should be able to **register and login** to the application.
2. Users should be able to **create posts** (text / image / video).
3. Users should be able to **follow other users** or send friend requests.
4. Users should be able to **like and comment** on posts.
5. Users should be able to **view a feed** consisting of posts from users they follow.



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Non-functional Requirements

A. Scalability

- System should support up to 500 million daily active users (DAU).

B. Consistency and Availability

- The system prioritizes high availability over strong consistency.
- Temporary delays in post propagation are acceptable.

Justification:

If the application is unavailable during peak time, it leads to a major business loss.

Example:

If Instagram is down for 1 hour → **critical issue**

If a post reaches followers in 500 ms instead of instantly → **acceptable** Hence,

Availability >>> Consistency

C. Latency

- Post upload and publish latency should be around **500 ms**.

Outcome / Result -

A complete high-level design of a social media platform was successfully created, identifying its functional requirements, non-functional constraints, core entities, and feed management strategy.

Users

- userID
- Username
- Email
- Password
- Phonenumber
- Followers_count
- friends_count
- profile_url
- other meta data of user

Post (Document DB)

- post_id
- user_id
- post_type
- content_text
- media_url
- thumbnail_url
- like_count
- share_count
- comment_count
- other meta data

Registration: Data Saving in DB

User DB

PostgreSQL

POST DB

Write Ops Fast

Amazon S3

API Gateway & Load Balancers

- Authentication
- Authorization
- Routing
- Rate Limiting

Clients

User Service

Content Service

Post Materializer

KAFKA Producer

Post Consumer SVC

Fanout Svc (PUSH)

Fanout Consumer

Feed Service

Feed DB

Cache

BackFill

1. User Registration

2. Logging

JWT for session management

Check user's credentials

HTTP Response: User Verified

Post against policy

Notification SVC

Moderator SVC

Images

Videos

For normal person

Users Latest post < recent 100 posts pre-computed >

< userID, List <postID>

< post, List<Friends/UserId>

3. Get me all recent post based on my followers

For a specific user storing all post's to show on feed.

3. write

3. write

1. Will Check for the followers from follower DB

Feed Cache Feed preparation in cache

Follower Cache (Top Followers)

